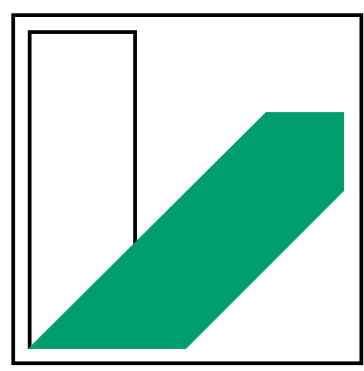


Mosses and *Nostoc*: an update from the Fichtel Mountains

Bayreuther Zentrum für
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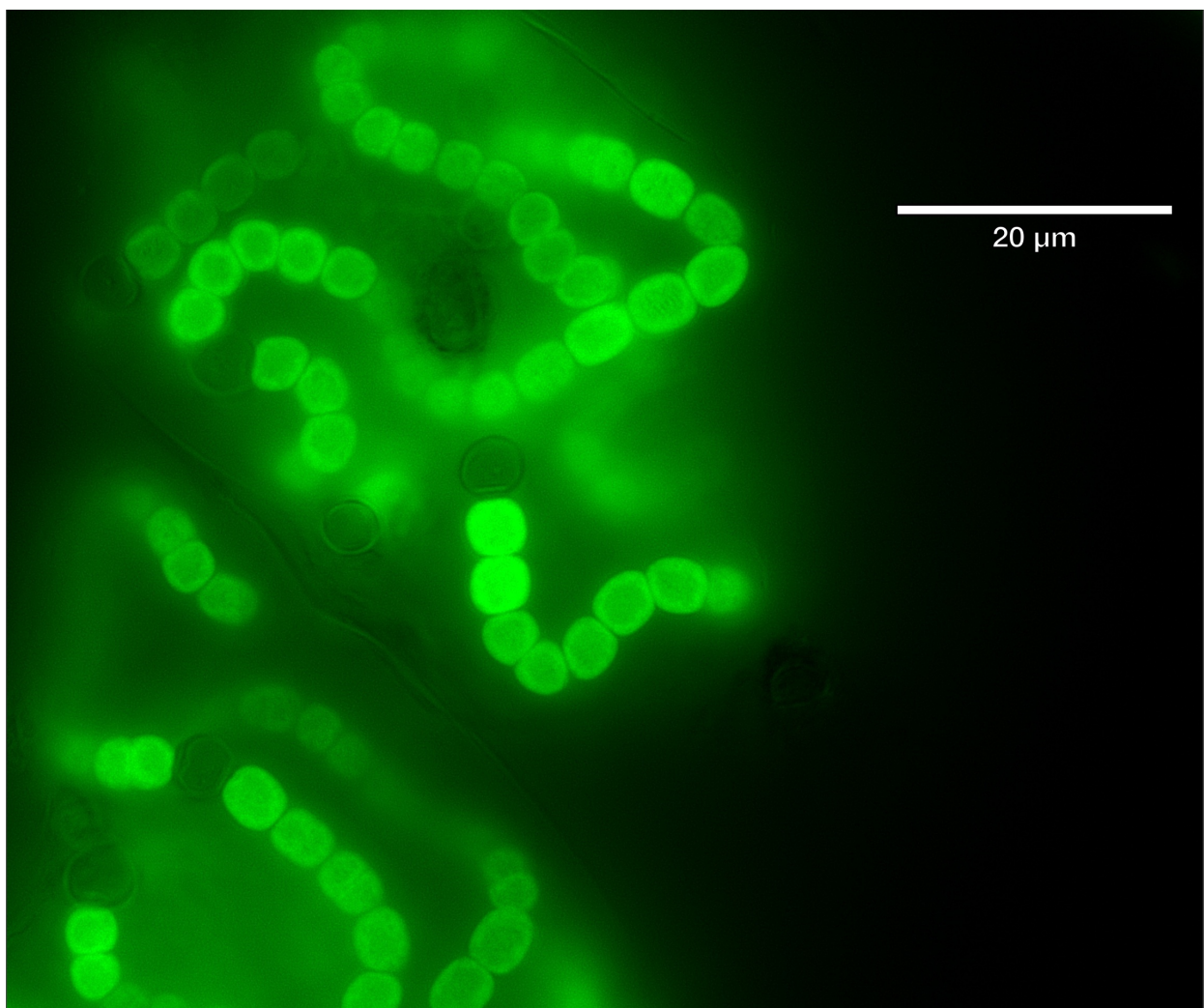
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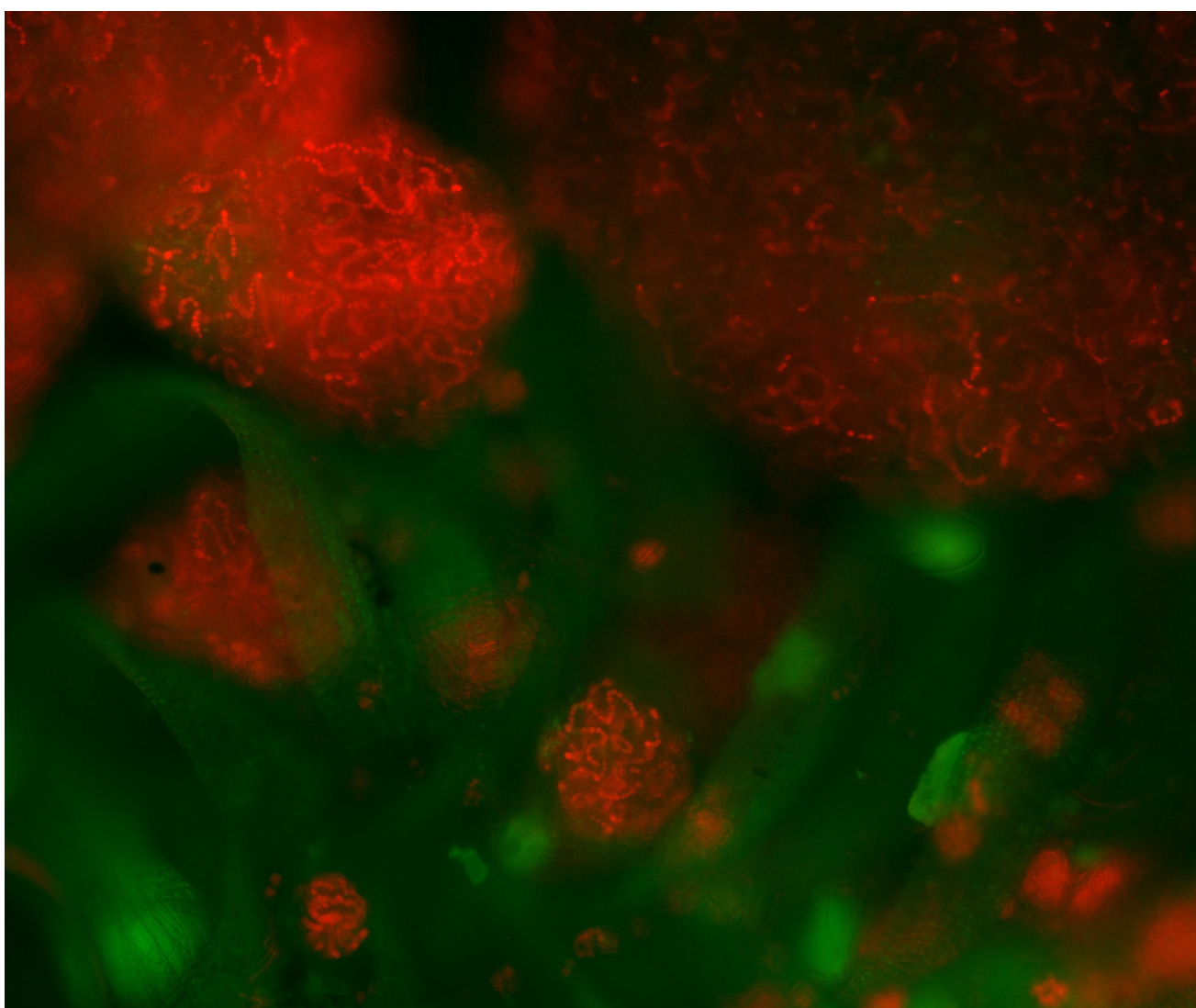
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Introduction

- Mosses are hotspots of nitrogen-fixation, due to diazotrophic symbionts in their microbiomes.
- Because of the large scale ecological service provided by these microbes, moss-cyanobacterial symbioses are of increasing interest.
- Historically the most commonly studied such symbioses are *Nostoc* spp. with "feather" (Hypnales) and *Sphagnum* mosses.
- Most studies have occurred in Scandinavia and the North American Boreal forest zone.



Nostoc sp. cultured from *Hylocomium* moss, forest floor. Epi-fluorescence microscope with Chroma FITC filter settings (CHROMA 49002, ex: 450-490 nm, em: 500-550 nm).



Compactonostoc sp. on *Zygodon dentatus* moss. Epi-fluorescence microscope with Chroma FITC filter settings (CHROMA 49002, ex: 450-490 nm, em: 500-550 nm) for moss tissue and Cy5 filter settings (CHROMA 49009, ex: 610-650 nm, em: 660-735 nm).



Waldstein forest, a typical field site in the Fichtengebirge mountains.



Hylocomium splendens moss with *Nostoc* colonies. Drawing by Jona Motzkus (2024).

Hypotheses

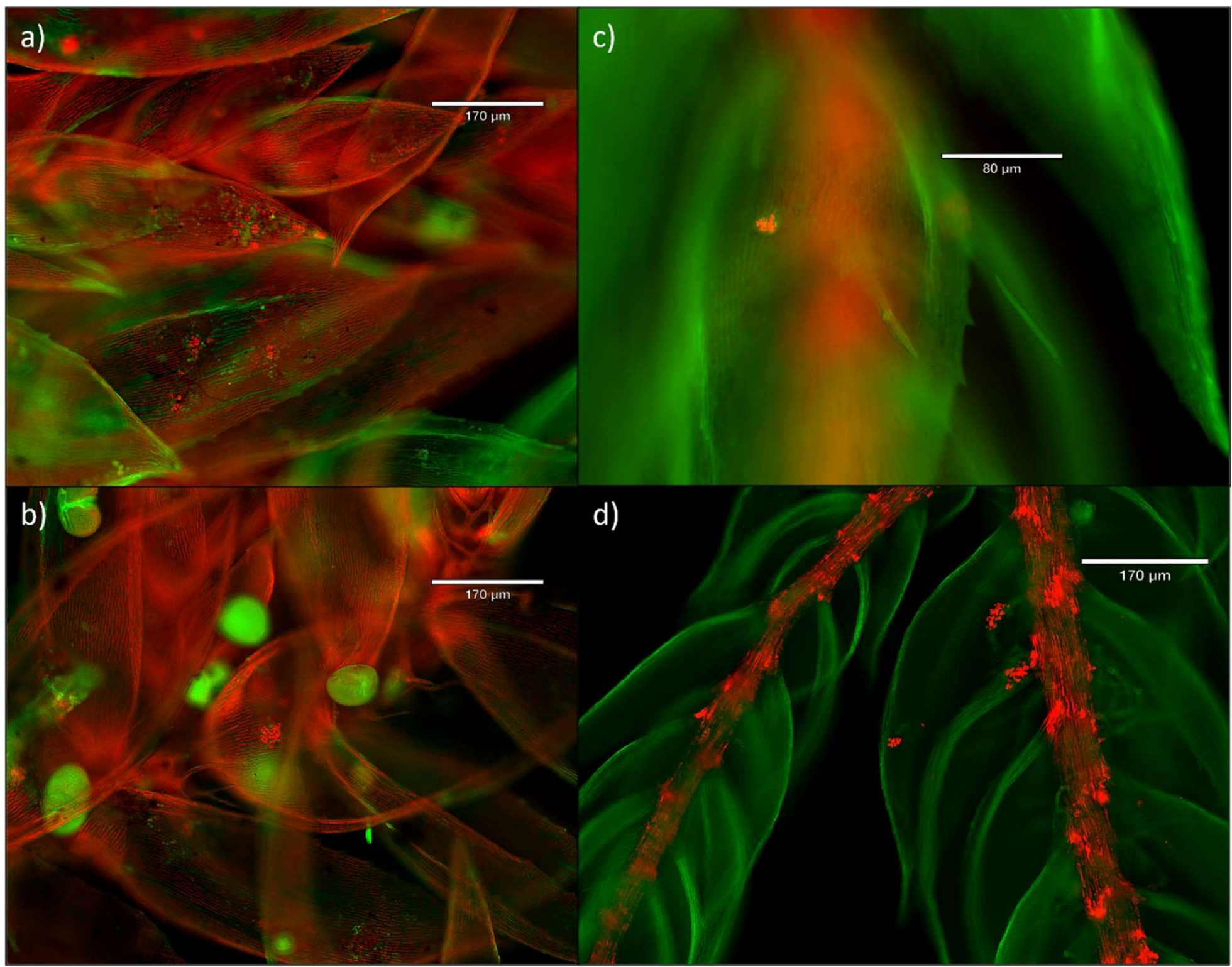
- Moss-*Nostoc* symbioses are "loose", better predicted by environmental conditions than host moss phylogeny.
- Mosses in the forest canopy are under selection pressure to enter into symbioses with nitrogen-fixing bacteria due to ephemeral environmental nitrogen sources.

Methods

- Field surveys for *Nostoc*-like colonies in mosses, especially rare arboreal mosses
- Light and epi-fluorescence microscopy
- attempted culturing of all *Nostoc*-like organisms
- chemical surface sterilization for detection of endophytes
- colony PCR of *NifH* and full-length 16S
- nanopore sequencing

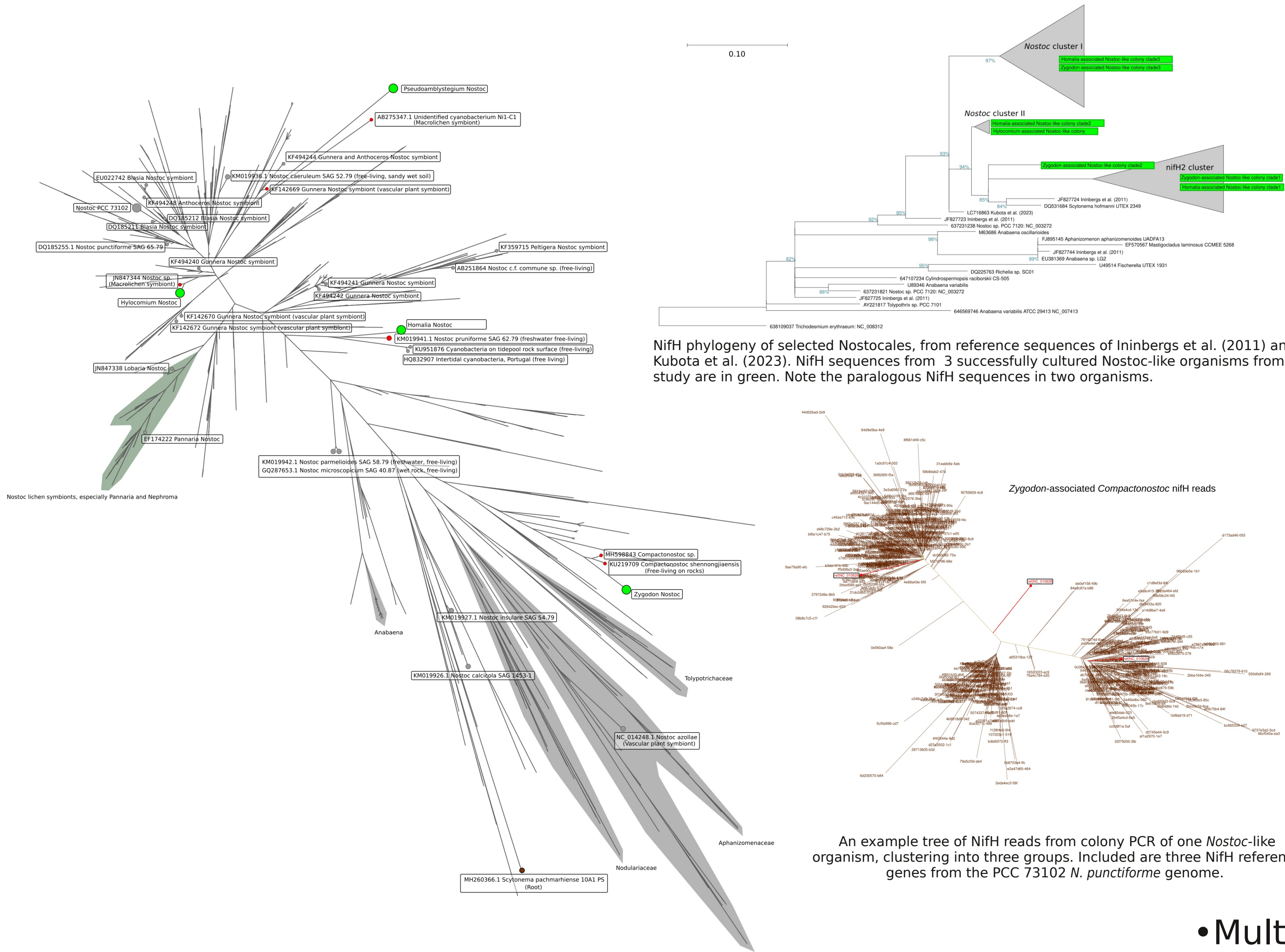


Culturing of cyanobacteria spp. cultured from moss frond in BG-11, liquid media.



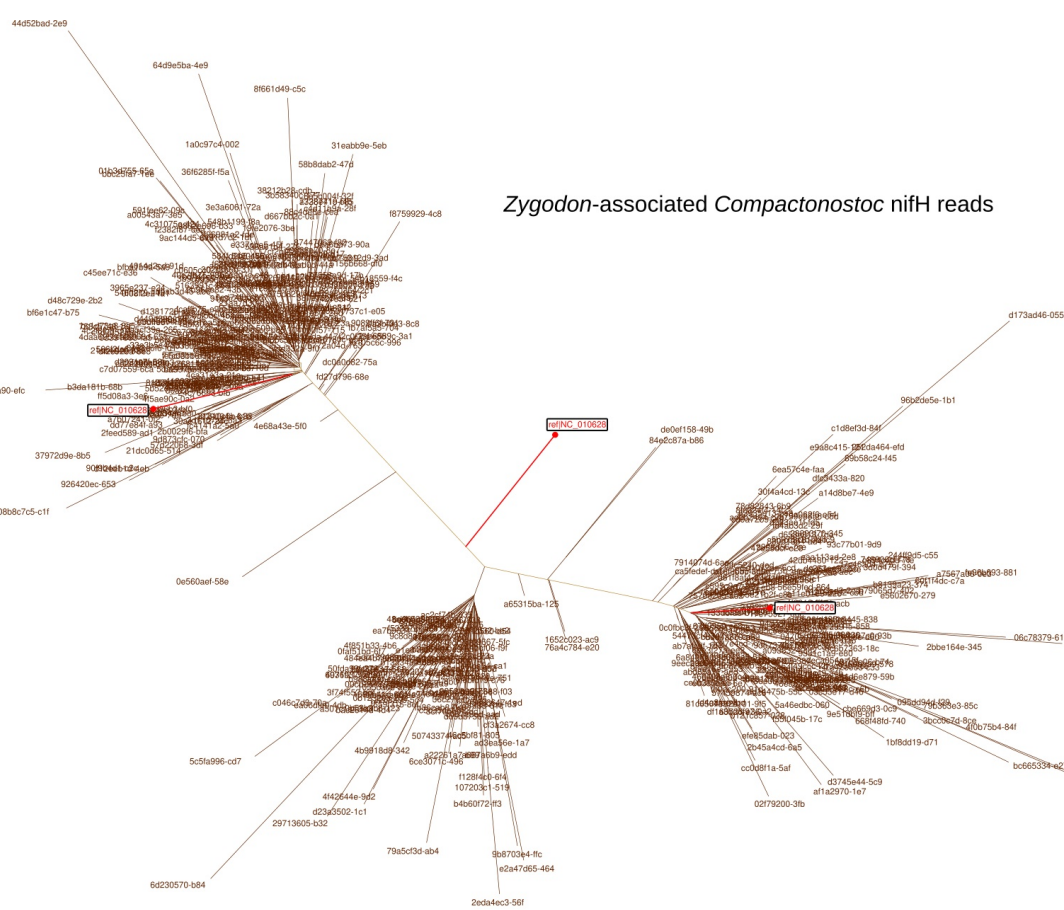
Surface sterilization of moss leaves: (a) and (b) are before surface sterilization, with extensive colonization by cyanobacteria and eukaryotic algae; (c) and (d) are post sterilization, with most foliar phototrophs removed but with colonies still apparent on stem tissue. Epi-fluorescence settings as per images above.

Results



Phylogeny of Nostocaceae and allied groups. Tree is constructed denovo from curated, full-length 16S sequences from the CyanoSeq database v1.3 (github.com/fleffler/CyanoSeq). Successfully amplified 16S sequences from cultures in this study are green, nearest neighbor sister taxa to these are in red.

NifH phylogeny of selected *Nostocales*, from reference sequences of Ininbergs et al. (2011) and Kubota et al. (2023). *NifH* sequences from 3 successfully cultured *Nostoc*-like organisms from our study are in green. Note the paralogous *NifH* sequences in two organisms.



An example tree of *NifH* reads from colony PCR of one *Nostoc*-like organism, clustering into three groups. Included are three *NifH* reference genes from the PCC 73102 *N. punctiforme* genome.

- Possible endophytic colonization by cyanobacteria of stems of mosses was observed.
- A new moss host family was observed (Orthotrichaceae) extending the range of known moss hosts for *Nostoc*-like organisms beyond feather mosses and *Sphagnum*.
- Arboreal mosses *Orthotrichum* and *Zygodon* (Orthotrichaceae) were frequent hosts to *Nostoc*-like organisms, possibly indicating a specialized niche for the symbiosis.

- Multiple alternate *nifH* genes were observed within the genomes of some *Nostoc*-like organisms, possibly enabling diazotrophy under variable ecological conditions.

Ininbergs, K., Bay, G., Rasmussen, U., Wardle, D. A., & Nilsson, M.-C. (2011). Composition and diversity of *nifH* genes of nitrogen-fixing cyanobacteria associated with boreal forest feather mosses. *New Phytologist*, 192(2), 507-517. <https://doi.org/10.1111/j.1469-8137.2011.03809.x>

Kubota, M., Matsushita, N., Nakamura, T., & Fukuda, K. (2023). Nitrogen fixation and *nifH* gene diversity in cyanobacteria living on feather mosses in a subalpine forest of Mt. Fuji. *Oecologia*, 201(3), 749-760. <https://doi.org/10.1007/s00442-023-05334-9>

- moss-associated *Nostoc*-like organisms were positioned throughout our 16S Nostocaceae phylogeny, with no obvious single moss-symbiotic clade in the family.